

SilverPaws AI

Internet video test report

A clear, constructive review of what the first MoveNet Thunder tests can tell us today.

Bottom line: the pipeline runs successfully, but the model is not yet reliable enough to make pet-specific gait claims from every video. One clip produced no usable ankle signal; the other produced a strong symmetry score, but the model may have been following a nearby human.

What was tested

Two publicly available dog videos were downloaded temporarily and analyzed outside the beta web app. Each test used the same MoveNet Thunder model, the same 0.20 confidence threshold, and the same first 120 frames so the comparison is fair.

Test	Confident ankle frames	Symmetry	Meaning
Test 1: dog + human walking	0 left / 0 right	Unavailable	No usable gait signal
Test 2: trained dogs hurdle clip	117 left / 120 right	98.4/100	Strong signal, needs subject validation

Test 1: dog and human walking clip

Measurement: 120 frames at 25 frames per second. Average confidence across all keypoints was 0.039. Neither ankle reached the confidence threshold in any frame.

Stride symmetry score

Unavailable

No score was reported because there was not enough confident ankle data.

Constructive reading: this is a useful failure, not a bad result. The validator correctly refused to turn weak detections into a confident number. The next improvement is subject selection: the clip contains both dogs and people, so a single-person model has an ambiguous target.

Test 2: trained dogs hurdle clip

Measurement: 120 frames at 25 frames per second. Average confidence across all keypoints was 0.405. The left ankle was confident in 117 frames and the right ankle in 120 frames.

Stride symmetry score

98.4%



This is a motion-similarity measurement, not a diagnosis.

Estimated asymmetry

1.6%



Lower is more similar within this clip.

Constructive reading: the pipeline can produce a stable, explainable number when the detections are strong. However, the result is not proof that the dog was tracked: MoveNet Thunder is trained on human pose, and the historical clip includes people. Treat this as a pipeline success and a model-validation warning at the same time.

How the number is calculated

For each frame, the validator reads the normalized vertical position of the detected left and right ankle. It adds the frame-to-frame movement for each side, then compares the two totals:

$$\text{symmetry} = 1 - |\text{left motion} - \text{right motion}| / (\text{left motion} + \text{right motion})$$

A score near 1.0 means the two detected sides moved similarly in that clip. It does not identify the cause of a movement pattern, and it cannot say whether a pet is healthy or injured.

What to do next

- 1. Use pet-specific pose detection.** Replace or supplement MoveNet Thunder with a model trained on dogs and cats, or add a subject-detection and cropping step before pose estimation.
- 2. Build a small labeled test set.** Collect at least 20 to 30 short clips with different breeds, camera angles, lighting, and walking surfaces. Mark whether the intended pet is actually detected.
- 3. Keep rejecting weak input.** The current validator already handles this correctly: if there are not enough confident ankle frames, it reports the metric as unavailable instead of inventing a reassuring score.
- 4. Compare a pet with itself.** For a useful wellness trend, repeat recordings with the same camera position and compare a pet's current observations with its own baseline.
- 5. Keep the language observational.** Say "the detected motion was less symmetric in this clip," not "your pet has arthritis." A veterinarian should interpret concerning changes.

Sources and limitations

Test 1: dog + human walking

Source: Wikimedia Commons file: Interspecific behavioural synchronization

License: CC BY 4.0 (Duranton, Bedossa, Gaunet)

Link: https://commons.wikimedia.org/wiki/File:Interspecific-behavioural-synchronization-dogs-exhibit-locomotor-synchrony-with-humans-41598_2017_12577_MOESM2_ESM.ogv

Test 2: trained dogs hurdle clip

Source: Wikimedia Commons file: Hurdle Jumping by Trained Dogs (1899)

License: Public domain historical film

Link: [https://commons.wikimedia.org/wiki/File:Hurdle_Jumping_by_Trained_Dogs_-_1899_-_American_Mutoscope_and_Biograph_\(US\)_-_EYE_FLM75062_-_OB_685436.ogv](https://commons.wikimedia.org/wiki/File:Hurdle_Jumping_by_Trained_Dogs_-_1899_-_American_Mutoscope_and_Biograph_(US)_-_EYE_FLM75062_-_OB_685436.ogv)

Important limitations

This report evaluates an engineering prototype. MoveNet Thunder is a pretrained human-pose model, not a validated veterinary or animal-pose model. Public videos may include people, occlusion, unusual camera angles, and uncontrolled motion. The score is not a medical diagnosis, emergency assessment, or substitute for a veterinarian.

If a pet shows pain, sudden weakness, collapse, breathing trouble, or another urgent change, contact a qualified veterinarian or emergency clinic directly. Do not wait for an app result.

SilverPaws AI beta | Experimental educational software | Generated from standalone internet-video tests